TT)-516d
August 20, 1962

SUPERSEDING
Int. Fed. Spec. TT-A-00516c (GSA-FSS)
February 15, 1962 and
Fed. Spec. TT-A-516b
April 7, 1961

FEDERAL SPECIFICATION

AMYL ALCOHOL: SECONDARY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. Secondary amyl alcohol covered by this specification is intended to be used in organic protective coatings.

1.2 Classification.

1.2.1 *Grade and type.* Secondary amyl alcohol covered by this specification shall be of one type and one grade (98 percent).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

TT-P-143—Paint, Varnish, Lacquer, and Related Materials; General Specification for Packaging, Packing, and Marking.

PPP-B-585–Boxes, Wood, Wirebound.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies)

Fed. Std. No. 141—Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Dever, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications, form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Code of Federal Regulations:

49 CFR 71-90—Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (Revised 1956) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-90 (Rev. 1956).")

ASTM Standards—Part 4

(Copies may be obtained from the American Society for Testing and Materials 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Quantitative requirements. The secondary amyl alcohol shall meet the requirements specified in table I.

3.2 Qualitative requirements.

3.2.1 *Appearance.* The amyl alcohol shall be clear and free from sediment and suspended matter when examined by transmitted light.

3.2.2 *Odor.* The material shall have the characteristic odor of secondary amyl alcohol and shall leave no residual odor after drying from filter paper for two hours.

3.2.3 *Water.* The amyl alcohol shall be miscible without turbidity with 19 volumes of 10° heptane.

TABLE I. *Quantitative requirements*

Characteristics	Requirements	
	Minimum	Maximum
Specific gravity at 20°/20° C.[1]	0.810	0.820
Nonvolatile matter from 100 ml., gram	——	0.005
Acidity, mg. of KOH per gram of sample	——	0.06
Distillation:		
Initial boiling point at 760 mm., pressure	113°C.	——
Distillate below 114° C., percent by volume	——	5
Distillate below 120°C., percent by volume	95	——
Dry point at 760 mm. pressure, °C.	——	125°C.
Color (Platinum-Cobalt Scale)	——	20
Flash point, °F.	100	110

[1](See par. 6.4)

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection. Sampling and inspection shall be in accordance with methods 1011, 1021, and 1031 of Fed. Test Method Std. No. 141, as applicable.

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4.2.1 *Test.* The following tests shall be in accordance with Fed. Test Method Std. No. 141:

Test	Method
Appearance	4261
Acidity	5252
Specific gravity	4183
Water	4083
Flash point	4294
Distillation[1] [2]	4301
Odor	4401

[1]Thermometer used shall be an ASTM solvents distillation thermometer conforming to the requirements of ASTM designation E 1 (41C 52).

[2]"Dry point" is the temperature observed on the distillation thermometer when the last drop of liquid leaves the bottom of the flask.

4.2.2 *Nonvolatile matter.* Place 100 milliliters of the sample in a weighed porcelain dish and evaporate over a steam bath or hot plate almost to dryness. Heat the dish and residue in an oven at 105° ± 2°C. (221° ± 4°F.) to constant weight, cool in a desiccator, and weigh. Report the difference in weight as grams of nonvolatile matter per 100 milliliters of the sample.

4.2.3 *Color.* The color secondary amyl alcohol shall be determined by ASTM Method D 1209.

5. PREPARATION FOR DELIVERY

The definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging and packing: Secondary Amyl Alcohol furnished in 1-quart or 1-gallon cans, 5-gallon drums, 55-gallon drums shall be packaged and packed in accordance with TT-P-143 as specified for the applicable level.

5.2 Marking.

5.2.1 *Precautionary marking.* Each container shall be marked with a white label, with the following information printed thereon in red letters:

"WARNING: FLAMMABLE
Keep away from heat and open flame.
Keep container closed.
Use with adequate ventilation.
Avoid prolonged breathing of vapor.
Avoid prolonged or repeated contact with skin."

5.2.2 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with the requirements of Fed. Std. No. 123.

5.2.3 *Military agencies.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Secondary amyl alcohol covered by this specification is intended for use in the manufacture of organic coatings such as lacquers and dopes.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify packaging and packing containers.

6.3 Secondary amyl alcohol should be purchased by volume, the unit being a U. S. gallon of 231 cubic inches at 20°C. (68°F). The volume of deliveries may be determined by dividing the net weight, in pounds, by the weight per gallon. To obtain the weight per gallon, multiply the specific gravity at 20°/20°C. by 8.322. One gallon of secondary amyl alcohol at 20°C. (68°F). weighs between 6.74 and 6.84 pounds.

6.4 As a matter of information, the specific gravity range of material conforming to this specification (0.810 to 0.820) at 20°/20°C. is approximately equivalent to a range of 0.814 to 0.824 at 15.56°C. and a range of 0.806 to 0.816 at 25°/25°C.

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6.5 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Amyl alcohol.
Carload minimum weight 30,000 pounds.

Motor:

Amyl alcohol.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Navy—Sh

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